Code No: 156EA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August - 2022
RENEWABLE ENERGY SOURCES
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) With neat and necessary diagram, explain the working principle of Multiple-Blade Turbines.
b) Discuss in detail about demand side Management Options of renewable energy systems. [8+7]

2.a) Classify different types of winds and their effects of the Wind on Nature.
b) Derive the expression for wind power per swept area is equal to $0.25V^3$. [8+7]

3.a) Along with the constructional features, explain principle of operation of Solid Oxide Fuel Cell.
b) Write short notes on Practical determination of the equivalent model parameters of fuel cells. [8+7]

4.a) Discuss in detail about the practical issues related to Fuel Cell Stacking.
b) Draw the Equivalent model of a PV panel with identical cells in series and discuss. [8+7]

5.a) Explain the principle of operation of Induction Generator.
b) Write short notes on the mathematical description of the self-excitation process of IG. [8+7]

6. Discuss about the following Speed and Voltage Control of Induction Generators:
a) Variable-Speed Grid Connection
b) Control by the Load Versus Control by the Source. [7+8]

7.a) With neat diagram, explain the Constructional Features of the Lead–Acid Batteries and Battery Charge–Discharge Cycles.
b) Discuss in detail about Double-Layer Ultra capacitors and High-Energy Ultra capacitors. [7+8]

8.a) Draw the Interconnection system, functional schematic diagram of multiple RES and explain in detail about each component in it.
b) Discuss about the issues that should be examined to ensure that the area electrical power system and alternative energy systems integrate in a safe and reliable manner. [8+7]

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